What Makes Audio Latents Mixing-Equivariant? A Controlled Study of Explicit Supervision

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***Abstract*—Mixing is a linear operation on waveforms, so a latent space that respects it, one in which interpolating two codes and decoding reproduces the waveform mix, is an *equivariant* representation. Neural audio autoencoders do not learn this structure on their own. Recent work induces it in a consistency autoencoder by decoding averaged latents against the true mix; we ask which part of such supervision is actually responsible, whether the decoder class limits what it delivers, and whether the metrics used to certify it measure it at all. We answer with a controlled study in which each experiment intervenes on a single factor while holding architecture, data, initialization, and schedule fixed. An explicit *decode-mixing loss*, which requires decoded latent interpolations to match true waveform mixes, at the cost of one extra decoder pass, is sufficient: on a waveform GAN autoencoder it makes the latent space mixing-equivariant *at no reconstruction cost*, and latent subtraction removes a stem at $+5.7$ dB SI-SDR versus $+3.4$ dB without it. At $2\times$ tighter compression, against a matched from-scratch control, the loss is what makes the operation viable at all ($+1.2 \to +5.2$ dB). The interventions also identify what is *not* responsible: an adversarial term on the mixed path inflates a decode-consistency metric without improving ground-truth equivariance, exposing that metric as a confounded proxy; an encoder-only penalty linearizes the encoder without making the decoder equivariant; and on a consistency-model autoencoder (Music2Latent) a continued-training control rules out domain adaptation. Finally, the widely used decode-vs-decode linearity metric is shown to be confounded by decoder phase variance, and a ground-truth-referenced variant is reported throughout.**

***Index Terms*—representation learning, equivariant representations, audio autoencoders, latent arithmetic, controlled ablation, source separation**

I. Introduction

Audio mixing is a linear operation: $\mathrm{mix}(x_1, x_2, \alpha) = \alpha x_1 + (1-\alpha)x_2$. For an autoencoder with encoder f and decoder g, we call the latent space *mixing-equivariant* when

$$g(\alpha f(x_1) + (1-\alpha) f(x_2)) \approx \alpha x_1 + (1-\alpha) x_2. \qquad (1)$$

This property turns latent vector arithmetic into audio editing: subtracting a stem's latent removes it from a mixture, adding latents mixes sources, and weighted combinations remix, all without a dedicated separation or synthesis model. It is also a prerequisite for latent-diffusion systems [1], [2] that compose independently modeled sources. Yet standard autoencoders do not satisfy Eq. 1: their objectives reward reconstruction fidelity, not algebraic structure, and quantized codecs (DAC [3], EnCodec [4]) cannot represent interpolated codes at all.

Torres et al. [5] induce linearity in a consistency autoencoder by decoding the *average* of separately encoded latents and training it to reconstruct the true mix, a supervision principle related to mixup [6]. This leaves open which part of that supervision is responsible for the property, whether the encoder or the decoder must carry it, whether a consistency decoder can turn latent linearity into usable waveform arithmetic, and whether the metrics used to certify it measure it at all. We answer these questions with a controlled study: the same principle written as an explicit loss (Section III) on a waveform GAN autoencoder, whose effect we then isolate by intervening on one factor at a time. Our contributions are:

1) A controlled study of mixed-latent supervision, instantiated as an explicit decode-mixing loss on a waveform GAN autoencoder, where it yields mixing-equivariance *without degrading reconstruction* (Section IV-A) and phase-coherent latent-arithmetic stem removal (Section IV-C), which a consistency decoder does not deliver [5].
2) An interventional ablation isolating the mechanism: the decode-mixing loss alone is sufficient; adding an adversarial term on the mixed path inflates a decode-consistency metric with no ground-truth gain, exposing the metric as a confounded proxy; an encoder-only penalty linearizes the encoder without making the decoder equivariant; and freezing the encoder retains most of the gain, so the effect is carried through the decoder and is largest when both networks adapt.
3) Evidence that the recipe generalizes, each against a matched control: from-scratch training at $2\times$ tighter compression, where the loss is what makes latent subtraction viable at all, and transfer to a consistency-model autoencoder with a continued-training control ruling out domain adaptation. The gain holds on all three test domains.
4) A methodological correction: the widely used decode-vs-decode $\text{SI-SDR}_{\text{lin}}$ metric is confounded by decoder phase variance; we report a ground-truth-referenced variant.

II. Related Work

Neural audio codecs. DAC [3], EnCodec [4], and SoundStream [7] pair convolutional encoders with HiFi-GAN decoders [8] and multi-scale discriminators, optimizing reconstruction without latent structure; their residual vector quantization also precludes latent interpolation. RAVE [9] learns a variational latent space but does not target mixing equivariance, and disentanglement regularizers [10] encourage generic smoothness rather than a specific algebraic identity. **Equivariant representations.** Mixing-equivariance is an instance of the symmetry-based view of representation learning, in which a representation is characterized by how it transforms under a known group action on the data [11]; Verma and Berger [12] build equivariance to audio transformations into the architecture. We instead treat equivariance to the linear mixing operation as a property to be *induced* by supervision, and isolate by intervention which supervision induces it. **Linear audio representations.** Music2Latent [13] is a consistency autoencoder with partial linearity as a byproduct. Torres et al. [5] make it approximately linear by conditioning the decoder on the average of separately encoded latents while denoising the true mix under the unchanged consistency loss (plus a random gain for homogeneity, which we do not study). Our $\mathcal{L}_{\text{mix}}^{\text{dec}}$ is the explicit-loss form of that additivity supervision: the same input (averaged latents) and target (true mix), differing in loss family (multi-resolution STFT and mel versus consistency), mixing coefficient ($\alpha \sim U[0,1]$ versus 0.5), and decoder class (deterministic GAN versus consistency model). We add the analysis: which ingredient carries the effect, a matched re-run of the principle on their architecture with a continued-training control (Section IV-B), and ground-truth-referenced metrics. Their decoder-additivity and separation scores compare two decodes (decoded latent sum versus the *autoencoded* mix or source), the protocol Section III-B shows to be confounded, and their latent-arithmetic separation is negative on every stem (-0.3 to -1.8 dB SI-SDR), as our phase-coherence finding predicts.

III. Method

A. Decode-mixing loss

Given a batch $\{x_i\}$ we form pairs (x_i, x_j) with a fixed-point-free permutation and draw $\alpha \sim U[0,1]$. Let $\bar{x} = \alpha x_i + (1-\alpha)x_j$ be the waveform mix and $\bar{z} = \alpha f(x_i) + (1-\alpha) f(x_j)$ the latent interpolation. The decode-mixing loss requires the decoded interpolation to match the true mix:

$$\mathcal{L}_{\text{mix}}^{\text{dec}} = \mathcal{L}_{\text{recon}}(g(\bar{z}),\, \bar{x})\,, \qquad (2)$$

where $\mathcal{L}_{\text{recon}}$ is the multi-resolution STFT + mel loss also used for reconstruction. Gradients flow through g and back into f via $\bar{z}$, so a single term constrains both networks; it costs one extra decoder pass per step. We also study an *encoder-only* variant that penalizes encoder nonlinearity directly, $\mathcal{L}_{\text{mix}}^{\text{enc}} = \|f(\bar{x}) - \bar{z}\|^2$ (one extra encoder pass, no decoder pass). The total generator objective is $\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{GAN}} + \lambda\, \mathcal{L}_{\text{mix}}^{\text{dec}} + \lambda_{\ell_2} \|z\|^2$.

B. Metrics

We report reconstruction $\text{SI-SDR}_{\text{rec}} = \text{SISDR}(g(f(x)), x)$ and two equivariance views at α=0.5:

- **SI-SDR**$_{\text{lin}}$ (decode-vs-decode): $\text{SISDR}(g(\bar{z}), g(f(\bar{x})))$, equivariance relative to the model's *own* reconstruction of the mix.
- **SI-SDR**$_{\text{lin}}^{\text{gt}}$ (vs. ground truth): $\text{SISDR}(g(\bar{z}), \bar{x})$, the quantity in Eq. 1, comparable across models.

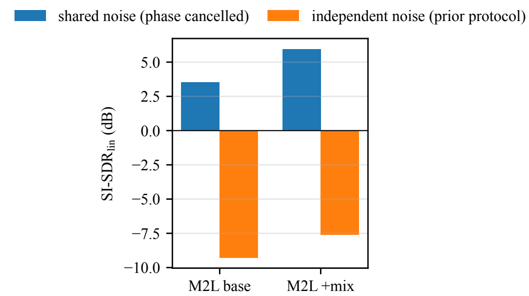

Fig. 1. Decode-noise protocol confound (Music2Latent, MUSDB18, α=0.5). The same checkpoints score deeply negative under independent-noise decoding and positive under shared-noise decoding; the ≈ -8 dB figure is a phase artifact, not a linearity measurement.

We also report $\ell_{\text{lat}} = \|\bar{z} - f(\bar{x})\|^2 / \|f(\bar{x})\|^2$ (encoder linearity) and $\text{MixRate} = \mathcal{L}_{\text{recon}}(g(\bar{z}), \bar{x}) / \mathcal{L}_{\text{recon}}(g(f(\bar{x})), \bar{x})$ (decoder equivariance; < 1 is better than the encode-then-decode path). SI-SDR$_{\text{lin}}$ is the metric used by prior work, but it is *gameable*: a decoder driven onto a common realistic manifold raises the mutual SI-SDR of its two decode paths without bringing either closer to $\bar{x}$ (Section IV-A). It is also confounded for stochastic decoders: a consistency-model decoder [13] draws fresh noise per call, so two decodes of the same latent differ in phase and SI-SDR$_{\text{lin}}$ collapses unless the noise is shared. Fig. 1 shows the same Music2Latent checkpoints scoring ≈ -8 dB under independent-noise decoding (the protocol implied by prior reports) and *positive* under shared-noise decoding, a 12 dB swing driven purely by the eval protocol, not the model. We therefore treat SI-SDR$_{\text{lin}}^{\text{gt}}$ and MixRate as the equivariance metrics of record, and use shared decode noise everywhere.

C. Experimental setup

The primary model is a waveform autoencoder (DAC-style 1-D convolutional encoder, HiFi-GAN decoder [8], combined multi-scale-STFT and multi-period discriminator) operating at 44.1 kHz on 1-second mono chunks, trained on a ~950-hour union of FMA-large, MAESTRO, and MUSDB18-HQ [14].[1] Two compression regimes are used: 7.66$\times$ (d=128) and 15.3$\times$ (d=64). Mixing losses are added either as a 25k-step fine-tune over a converged baseline (v2.x) or from scratch over 250k steps (v3.x). All runs use a single NVIDIA RTX 5090. Unless noted, metrics are computed over the full multi-source test set; the α-sweep (Section IV-D) uses a source-balanced subsample.

IV. Results

A. Which signal causes equivariance? An interventional ablation

Table I ablates the loss on the 7.66$\times$ GAN autoencoder (25k-step fine-tunes from a shared converged baseline). Each

[1]Code, training configurations, pretrained weights, every metric reported here, and audio examples: https://github.com/nurdauletakhanov/musicgen

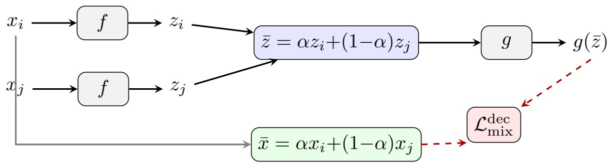

Fig. 2. Decode-mixing loss: encode a pair, interpolate latents, decode, and compare to the true waveform mix. Gradients (dashed) flow through the decoder and back into the encoder via $\bar{z}$.

Loss	$\mathrm{SDR_{rec}}$	$\mathrm{SDR_{lin}}$	$\mathrm{SDR^{gt}_{lin}}$	ℓ_{lat}	MixRate	FAD
baseline (no mix)	+11.3	+12.0	+8.0	0.068	1.203	0.045
$\mathcal{L}_{\mathrm{dec}}$	+11.4	+12.1	+10.2	0.052	0.972	0.044
$\mathcal{L}_{\mathrm{dec}}$ + disc-on-mix	+11.4	+16.0	+9.9	0.050	1.052	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=5)	+11.3	+12.8	+8.4	0.044	1.175	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=10)	+11.3	+13.1	+8.5	0.039	1.165	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=20)	+11.3	+13.5	+8.7	0.034	1.153	0.045
$\mathcal{L}_{\mathrm{dec}}$ frozen-enc	+11.2	+11.7	+9.5	0.072	1.056	0.046

TABLE I
MECHANISM ABLATION ON THE GAN AUTOENCODER (7.66×, FULL TEST SET, α=0.5). $\mathrm{SDR_{lin}}$ IS DECODE-VS-DECODE; $\mathrm{SDR^{gt}_{lin}}$ IS VS. GROUND TRUTH. SINGLE SEED.

row differs from its comparator in exactly one factor: $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ and $\mathcal{L}^{\mathrm{enc}}_{\mathrm{mix}}$ against the baseline (the loss), $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$+disc against $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ (the adversarial term), and frozen-encoder $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ against $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ (the encoder's freedom to adapt); the design is not a full factorial. Three observations drive the paper.

(1) Reconstruction is unaffected. $\mathrm{SDR_{rec}}$ is flat across all configurations (11.2–11.4 dB), and so is FAD (0.044–0.046): the mixing loss buys equivariance for free, on both a sample-wise and a distributional measure of quality.

(2) The decode-mixing loss is the active ingredient; the discriminator on the mixed path is not. On the ground-truth metric, $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ alone improves $\mathrm{SDR^{gt}_{lin}}$ from 8.0 to 10.2 dB and is the only run with MixRate < 1 (0.97). Adding an adversarial term to the mixed path ("+disc-on-mix") does *not* improve ground-truth equivariance (9.9 dB, MixRate 1.05), yet it dramatically inflates the decode-vs-decode $\mathrm{SDR_{lin}}$ (12.1 → 16.0 dB). The discriminator pulls the two decode paths onto a common realistic manifold, raising their mutual SI-SDR without moving either toward $\bar{x}$. This is direct evidence that $\mathrm{SDR_{lin}}$ is gameable, and that given $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ no adversarial term is needed; whether the discriminator has an effect on its own, without $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$, is not tested here.

(3) The effect is carried through the decoder. The encoder-only variant $\mathcal{L}^{\mathrm{enc}}_{\mathrm{mix}}$ drives ℓ_{lat} lowest but leaves MixRate high (1.15–1.18): it linearizes the encoder without making the decoder equivariant. Conversely, training only the decoder under $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ with the encoder frozen retains most of the gain (9.5 dB versus 10.2 joint and 8.0 baseline) while leaving encoder linearity unimproved (ℓ_{lat} 0.072). Joint training is best among the tested configurations.

B. Intervening on compression and architecture

Table II extends the recipe along two axes. **Compression:** trained *from scratch* at 15.3× (v3.1), the mixing model

Model	$\mathrm{SDR_{rec}}$	$\mathrm{SDR_{lin}}$	$\mathrm{SDR^{gt}_{lin}}$	MixRate	FAD
GAN AE, 7.66×, no mix	+11.3	+12.0	+8.0	1.203	0.045
GAN AE, 7.66×, +mix	+11.4	+12.1	+10.2	0.972	0.044
GAN AE, 15.3×, no mix	+10.0	+10.6	+6.8	1.187	0.052
GAN AE, 15.3×, +mix	+9.9	+14.6	+8.6	1.046	0.061
M2L, no mix (baseline)	-3.3	+5.0	-9.6	1.101	0.127
M2L, fine-tune control	-3.1	+5.1	-9.5	1.125	–
M2L, +mix	-2.6	+6.9	-7.8	1.086	0.119

TABLE II
GENERALIZATION ACROSS COMPRESSION AND ARCHITECTURE (FULL TEST SET, α=0.5, EMA WEIGHTS FOR MUSIC2LATENT). THE M2L "FINE-TUNE CONTROL" IS THE CONTINUED-TRAINING BASELINE WITH MIXING DISABLED.

$\mathrm{SDR^{gt}_{lin}}$	FMA	MAESTRO	MUSDB	all
7.66×, no mix	+6.8	+13.8	+6.2	+8.0
7.66×, +mix	+8.9	+16.9	+8.2	+10.2
Δ	**+2.1**	**+3.1**	**+2.0**	**+2.3**
15.3×, no mix	+5.8	+12.0	+5.5	+6.8
15.3×, +mix	+7.2	+15.1	+6.7	+8.6
Δ	**+1.4**	**+3.2**	**+1.2**	**+1.7**

TABLE III
GROUND-TRUTH EQUIVARIANCE PER TEST DOMAIN (DB). EACH Δ IS AGAINST THE MATCHED NO-MIXING CONTROL AT THE SAME COMPRESSION RATE. THE GAIN IS PRESENT ON EVERY DOMAIN, NOT CARRIED BY ONE.

matches its no-mixing control (v3.0) on reconstruction (9.9 vs. 10.0 dB) while improving ground-truth equivariance by +1.7 dB ($\mathrm{SDR^{gt}_{lin}}$ 6.8 → 8.6), MixRate (1.19 → 1.05), and encoder linearity (ℓ_{lat} 0.078 → 0.057). Because v3.0 is a matched control (same architecture, compression, and schedule, mixing disabled), this isolates the loss's contribution and shows the effect is not an artifact of the looser 7.66× rate or of warm-starting. **Architecture:** we fine-tune the published Music2Latent [13] consistency autoencoder with the decode-mixing objective, the additivity supervision of [5] on its own architecture, using its native consistency loss on the mixed pair in place of the discriminator; this is the matched comparison in which only the decoder class differs from the GAN autoencoder. A continued-training control, the same fine-tune with the mixing terms disabled, moves equivariance by < 0.2 dB and *worsens* FAD, so the mixing model's gain is attributable to the loss, not to domain adaptation. Both decode-vs-decode and ground-truth metrics agree on the improvement, confirming it is not a measurement artifact. Absolute $\mathrm{SDR^{gt}_{lin}}$ remains negative on Music2Latent because its consistency decoder is phase-incoherent with raw waveforms; the latent *geometry* improves (MixRate, ℓ_{lat}) but a consistency decoder does not convert this into clean waveform arithmetic as the GAN decoder does, so we report the GAN autoencoder as the primary result.

The gain is not domain-driven. Our test set spans three acoustically distinct domains: crowd-sourced indie production (FMA), solo piano (MAESTRO), and multi-track studio mixes (MUSDB18). Table III breaks $\mathrm{SDR^{gt}_{lin}}$ out by domain against the matched no-mixing control at each compression rate. The improvement appears on all three at both rates (+2.0 to +3.1 dB at 7.66×; +1.2 to +3.2 dB at 15.3×), so it reflects a property of the learned latent space rather than an artifact of one recording style. The absolute ordering (MAESTRO easiest, MUSDB hardest) tracks reconstruction difficulty and is preserved by the mixing loss.

C. Downstream: latent-arithmetic stem removal

We remove a stem by latent subtraction, $\hat{x} = g(f(x_{\mathrm{mix}}) - f(x_{\mathrm{stem}}))$, and measure SI-SDR against the true residual on the MUSDB18 test set. This tests *extrapolation* beyond the convex range $\alpha \in [0, 1]$ seen in training. Alongside the subtraction score we report the *linearity tax* gap $=$ $\mathrm{SISDR}(g(f(x_{\mathrm{res}})), x_{\mathrm{res}}) - \mathrm{SISDR}(\hat{x}, x_{\mathrm{res}})$, the shortfall of latent arithmetic against the same model's own encode–decode ceiling on the residual. The gap normalizes away raw reconstruction quality and so compares models at different compression rates on equal terms.

Table IV shows the mixing loss roughly doubles usable subtraction quality at 7.66× (+3.4 → +5.7 dB overall) and cuts the linearity tax from 5.2 to 3.0 dB, with the largest gain on bass (+0.4 → +4.9 dB). Adding disc-on-mix is comparable rather than better (+6.0 dB, tax 2.6), consistent with Section IV-A.

How much of this is measurement noise? Every model is scored on the same 5,880 (track, stem, chunk) units, so we compare them pairwise and bootstrap the mean difference (10^4 resamples). The gains sit far outside evaluation variance: $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ over the no-mixing control is +2.33 dB (95% CI [+2.26, +2.39], improving 77% of units), and at 15.3× the matched-control gain is +4.02 dB ([+3.93, +4.11], 90% of units). Disc-on-mix beats $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ alone by only +0.31 dB ([+0.29, +0.33]) here while scoring 0.36 dB *lower* on ground-truth equivariance (Table I): the two differ little, and in opposite directions depending on the metric. These intervals cover evaluation variance only, not variation across training seeds.

The effect is starkest against the matched from-scratch control at 15.3×: without the mixing loss, latent subtraction is essentially unusable (+1.2 dB overall, and −1.6 dB on bass, worse than emitting the unmodified mixture), while the same architecture trained with it reaches +5.2 dB and the *lowest* tax of any model (2.5 dB). The two settings differ in training history as well as compression (from scratch versus fine-tuned), so our design does not separate which of the two makes the control's arithmetic fail; what each matched pair isolates is that the loss restores it.

D. Equivariance across mixing ratios

Sweeping α on a source-balanced subsample (Fig. 3), the mixing loss *flattens* the equivariance curve: the no-mixing baseline sags most at the hardest balanced mix (α=0.5) and recovers toward the extremes, while mixing models stay near-uniform across α (e.g. $\mathrm{SDR}_{\mathrm{lin}}^{\mathrm{gt}}$ swing of 1.2 dB for $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$+disc vs. 3.2 dB for the baseline). Thus the property holds across mixing ratios, not only at the trained operating point.

Model	drums	bass	vocals	other	all	gap↓
7.66× no mix	+4.2	+0.4	+5.2	+3.9	+3.4	5.2
7.66× +$\mathcal{L}_{\mathrm{dec}}$	+5.7	+4.9	+7.2	+5.1	+5.7	3.0
7.66× +$\mathcal{L}_{\mathrm{dec}}$+disc	+6.2	+5.4	+7.5	+5.0	+6.0	2.6
15.3× no mix	+1.2	-1.6	+3.0	+2.1	+1.2	6.6
15.3× +mix	+5.4	+4.7	+6.9	+3.9	+5.2	2.5
M2L +mix	-9.9	-12.3	-9.9	-12.7	-11.2	8.4

TABLE IV
STEM REMOVAL VIA LATENT SUBTRACTION (SI-SDR, DB; MUSDB18 TEST). HIGHER IS BETTER; "GAP" IS THE LINEARITY TAX AGAINST EACH MODEL'S OWN ENCODE–DECODE CEILING (LOWER IS BETTER). PAIRED BOOTSTRAP CIS FOR THE MATCHED CONTRASTS ARE IN THE TEXT.

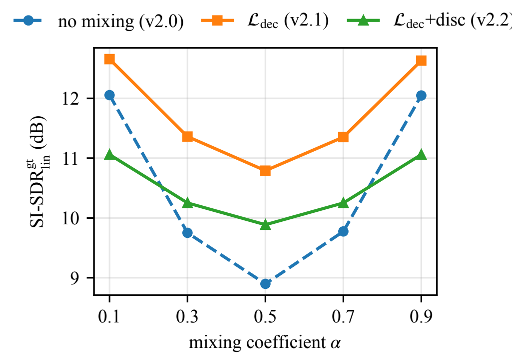

Fig. 3. Ground-truth equivariance vs. mixing coefficient α (source-balanced subsample). The no-mixing baseline sags at the hardest balanced mix (α=0.5); the decode-mixing loss flattens and lifts the curve across all α.

We caution that the decode-vs-decode $\mathrm{SDR}_{\mathrm{lin}}$ is U-shaped in α for *all* models (near-identity mixes are trivially easy), which is a property of that metric, distinct from the phase-variance confound discussed in Section III-B.

V. Conclusion

A controlled study identifies an explicit decode-mixing loss as the training signal responsible for mixing-equivariance in a waveform autoencoder, obtained at no reconstruction cost and enabling latent-arithmetic audio editing. The loss itself, not an adversarial term on the mixed path, is the active ingredient, and the effect transfers across compression rates and to a consistency-model autoencoder under a continued-training control. **Limitations.** All results use a single training seed. Our confidence intervals cover evaluation variance, not variation across initializations, so small differences (e.g. $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ vs. $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$+disc, 0.31 dB) remain within plausible seed variance and we report them as "comparable." Experiments are mono; consistency-decoder latent arithmetic remains phase-limited; and we conduct no listening test. Stereo, multi-seed estimates, and latent-diffusion composition are future work.

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